Vin Bhaskara

Applied AI Research Lead – LLMs and Foundation Models | vinbhaskara.github.io, bhaskara@cs.toronto.edu

IIT Silver Medalist | Vector AI Scholar | Prev: Samsung AI, University of Toronto | Citizenship: Canadian, Location: Montréal

SUMMARY

Product-driven AI research and engineering, with differentiated and measurable impact. $200M+ production impact across 13M customers at RBC, on-device denoising and super-resolution for Samsung Galaxy flagships. 12 papers, 3 patents, 300+ citations.

WORK EXPERIENCE

8+ years of full-time experience in AI and software engineering

ML Research Lead / Senior Research Engineer, Foundation Models and LLMs — **Borealis AI (RBC AI Research), Montréal**
Deep Learning for Capital Markets and Credit Modeling at Canada's Largest Bank — Aug 2022 – **Present**

- Led R&D on fine-tuned masked language models and agentic RAG over Bloomberg chat across multiple trading desks, delivering **$5MM+ per desk** annually and co-inventing a **diverse-retrieval RAG** method (US Patent Application 19/575,161)
- Co-led R&D for the flagship credit capacity product using **foundation models** trained on transaction data across **13 million customers**, driving ~**$200MM** in 5-year NIBT with improved fairness and interpretability (US Patent Application 19/091,539)
- Shipped two additional **explainable** neural credit risk scoring models adding **~$20MM** projected annual NIBT – recognized with RBC Quarterly Team Performance Award (2026)

Research Engineer, Computer Vision — **Samsung AI Centre, Toronto**
Deep Learning for Image Enhancement and Synthesis — Feb 2020 – Jun 2022

- Co-invented **unsupervised super-resolution** training-data construction (US Patent 12,210,587) and developed **on-device digital-zoom** super-resolution for Samsung Galaxy flagship cameras (ECCV 2020 workshop paper)
- Led research on GraN-GAN (WACV 2022 paper): a piecewise gradient normalization method for **stable GAN training**, applied to image synthesis tasks inside Samsung's imaging pipeline
- Led research on multi-frame alignment for burst photography using neural implicit representations and **self-supervised blind image denoising** for low-light **night mode** – recognized with "Samsung Research America Rockstar" performance award (2021)

Software Engineer 2 — **Broadcom Inc., India**
Machine Learning and Big Data for Malware Detection — Jul 2016 – Jul 2018

- Co-led a production **XGBoost model on Norton Anti-Virus** using Symantec's **big data file telemetry** that reduced previously missed malware detections by **> 60%** – recognized with Level 1 & 3 company-wide recognition awards
- Led research on image representations of dynamic run-time file behavior for GAN-based **proactive malware detection** and **software categorization** (arXiv:stat.ML/1807.07525)

EDUCATION

M.Sc. in **Applied Computing** (Deep Learning), Department of Computer Science, 4.0/4.0 (**A+**) — Sep 2018 – Dec 2019
University of Toronto, Downtown Toronto, Canada

- Received the **Vector Institute Scholarship in Artificial Intelligence (VSAI)** valued at $17,500 awarded to **66 scholars in Ontario**
- Research: Part-Based Single-Shot Object Detection for Computer Vision. Supervisors: Dr. Alex Levinshtein and Prof. Allan Jepson.

B.Tech. in **Engineering Physics** with Minor in **Electronics Engineering**, Department **Rank 1** — Jul 2012 – Jun 2016
Indian Institute of Technology (IIT), Guwahati

- **Institute Silver Medalist** for the best academic performance in the department among the graduating class of 2016 at IIT Guwahati
- Primary author of a foundational paper on Quantum Entanglement theory and a **visiting scholar** at the Institute for Quantum Computing (IQC), University of Waterloo, Canada

RELEVANT PUBLICATIONS

Citations: **300+**, h-index: **8** - Google Scholar

1. **Vin Bhaskara**, H. Wang. "Curiosity-Critic: Cumulative Prediction Error Improvement as a Tractable Intrinsic Reward for World Model Training," ICML 2026 Workshop on Epistemic Intelligence in Machine Learning — May 2026
2. A. Ma, L. Schell, **Vin Bhaskara**, L. Pishdad. "From token probabilities to calibrated confidence: An empirical study of mathematical question answering," ICML 2026 Workshop on Structured Probabilistic Inference & Generative Modeling — May 2026
3. **Vin Bhaskara***, T.A. Armstrong*, A. Jepson, A. Levinshtein. "GraN-GAN: Piecewise Gradient Normalization for Generative Adversarial Networks," WACV 2022 Conference (2022 IEEE Winter Conference on Applications in Computer Vision) — Jan 2022
4. **Vin Bhaskara***, H. Wang*, A. Levinshtein*, S. Tsogkas, A. Jepson. "Efficient Super-Resolution Using MobileNetV3," ECCV 2020 Workshop on Efficient Super-Resolution (2020 European Conference on Computer Vision Workshop) — Jan 2021

5. **Vin Bhaskara**, S. Tsogkas, K. Derpanis, A. Levinshtein. "Part-based Auxiliary Objectives with No Extra Labels for Robust Single-Shot Object Detection," dx.doi.org/10.13140/RG.2.2.10079.47521 (Preprint) Apr 2020
6. **Vin Bhaskara**, S. Desai. "Exploiting uncertainty of loss landscape for stochastic optimization," arXiv:cs.LG/1905.13200 (Preprint) May 2019
7. **Vin Bhaskara**, Y. Fu, S. Gowda. "Risk Prediction in the General Internal Medicine Ward at St. Michael's Hospital," dx.doi.org/10.13140/RG.2.2.27695.55205 (Preprint) Apr 2019
8. **Vin Bhaskara**, D. Bhattacharyya. "Emulating malware authors for proactive protection using GANs over a distributed image visualization of dynamic file behavior," arXiv:stat.ML/1807.07525 (Preprint) Jul 2018

(Denotes equal contribution)*

GRANTED PATENTS

1. H. Wang, X. Sun, **Vin Bhaskara**, S. Tsogkas, A. Jepson, A. Levinshtein. "Unsupervised Super-Resolution Training Data Construction," Samsung AI Centre Toronto, US Patent 12,210,587 Jan 2025

PATENT APPLICATIONS

1. L. Schell, **Vin Bhaskara**. "Systems, Methods, and Techniques for Performing Retrieval Augmented Generation (RAG) with a Diverse-Retrieval Method," Borealis AI (RBC Research), US Patent Application No. 19/575,161 (Filed Mar 23, 2026) Mar 2026
2. S.H. Hajimirsadeghi, M.O. Ahmed, E.J. Smith, **Vin Bhaskara**, *et al.* "Generating Credit Capacity Estimates," Borealis AI (RBC Research), US Patent Application No. 19/091,539 (Filed Mar 17, 2025) Mar 2025

ACADEMIC SERVICE

- **Conference Reviewer / Program Committee** for NeurIPS, CVPR, ICLR, ICML, and AAAI 2023 – Present

TECHNICAL SKILLS

- **Languages**: Python (expert), Java, C++, C
- **AI Frameworks**: PyTorch, JAX, Hugging Face (Transformers, PEFT, Datasets), vLLM
- **Vector Databases**: FAISS, pgvector
- **Data and MLOps Tooling**: SQL, NoSQL, Hadoop, HDFS, MapReduce, Pandas, Docker, Weights & Biases

AWARDS AND HONORS

- **Vector Institute** Scholarship in Artificial Intelligence (VSAI) valued at $17,500 – **1 of 66 scholars in Ontario** 2018
- **Institute Silver Medalist, IIT Guwahati** – best academic performance in Engineering Physics, class of 2016 2016
- Offered **AI Residency** at **Google X** (Mountain View) – declined 2019
- **Kaggle "Competitions Expert"** – Ranked Top 5% Global, 5 Medals 2017 – Present
- **Top 5%** (201st of 4,436 teams) in a solo submission, earning a **Kaggle Silver Medal** for predicting Nasdaq stock price movements on real market data 2024
- USEQIP Summer School at the Institute for Quantum Computing (IQC), Waterloo, Canada – **25 students selected globally** 2015

OTHER PUBLICATIONS

1. **Vin Bhaskara***, S.N. Swain*, P.K. Panigrahi. "Generalized Entanglement Measure for Continuous-Variable Systems," Physical Review A (PRA) 105, 052441 (2022), American Physical Society Journals May 2022
2. **Vin Bhaskara**, P.K. Panigrahi. "Generalized concurrence measure for faithful quantification of multiparticle pure state entanglement using Lagrange's identity and wedge product," Quantum Inf. Process. 16 (5), 118, Springer Journals Mar 2017
3. J. Flannery, G. Bappi, **Vin Bhaskara**, O. Alshehri, M. Bajcsy. "Implementing Bragg mirrors in a hollow-core photonic crystal fiber," Optical Materials Express 7 (4), 1198, Optical Society of America Journals Mar 2017
4. C.M. Haapamaki, J. Flannery, G. Bappi, R. Al-Maruf, **Vin Bhaskara**, O. Alshehri, T. Yoon, M. Bajcsy. "Mesoscale cavities in hollow-core waveguides for quantum optics with atomic ensembles," Nanophotonics 5 (1), De Gruyter Journals Sep 2016

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